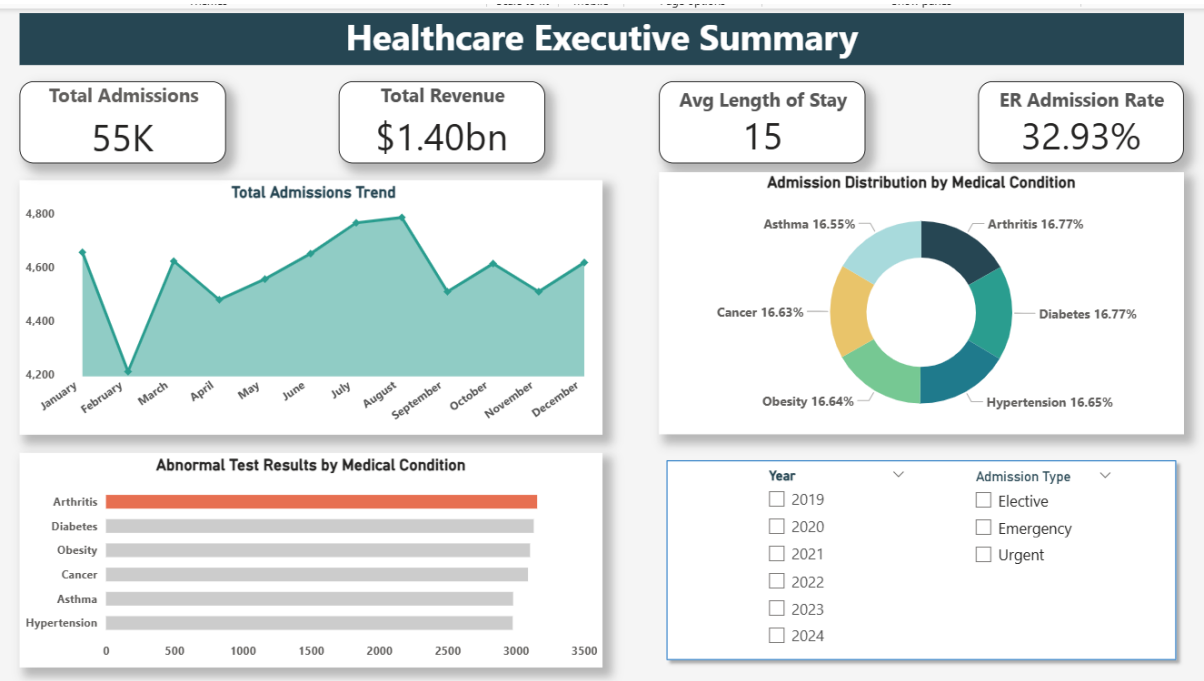
Project Implementation Steps

Project Objective

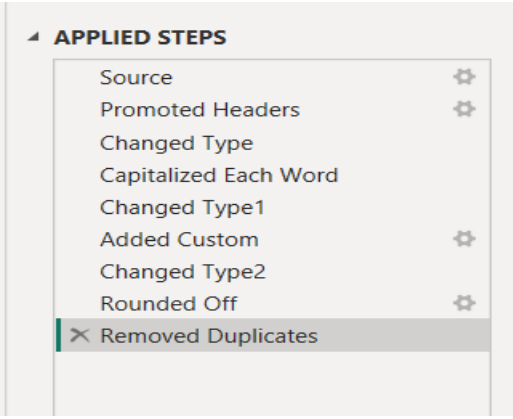
The goal of this project was to develop a comprehensive Healthcare Analytics solution to monitor hospital operations, patient outcomes, and financial performance. Using a dataset of 55,000 patient records, the dashboard provides actionable insights into billing trends, ER admission rates, and clinical risk factors. By centralizing data from multiple dimensions (Doctor, Hospital, Insurance, and Patient), this tool enables data-driven decision-making for hospital executives and clinical staff.



Data Transformation Process

Using Power Query, the raw healthcare data underwent the following transformations to ensure accuracy and consistency:

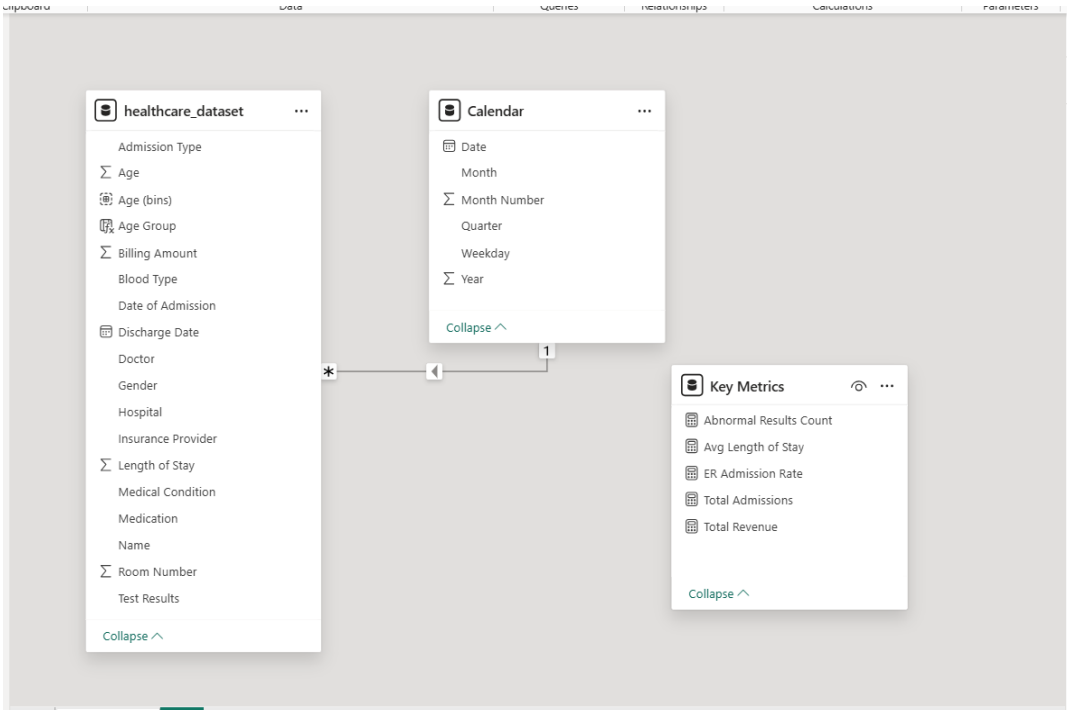
- **Standardization:** Applied 'Capitalize Each Word' to patient and doctor names to fix irregular casing.
- **Calculated Columns:** Added a custom column for 'Length of Stay' by calculating the difference between Discharge and Admission dates.
- **Data Quality:** Conducted 'Removed Duplicates' and ensured 'Changed Type' for dates and currencies to optimize data integrity.
- **Precision:** Rounded billing amounts to two decimal places for financial reporting clarity.



Data Architecture

The report is built on a robust **Star Schema** model, which is the industry standard for Power BI performance:

- **Fact Table:** healthcare_dataset serves as the central fact table containing clinical and financial records.
- **Dimension Tables:** A dedicated Calendar table was created to support Time-Intelligence analysis (Yearly/Monthly trends).
- **Calculation Groups:** A Key Metrics table was implemented to centralize all DAX measures for better maintainability.
- **Relationships:** Established a 1:Many relationship between the Calendar and healthcare_dataset tables.



Analytical Logic (DAX)

To drive the dashboard visuals, several complex DAX measures were authored:

- **Total Revenue:** SUM(Healthcare_dataset[Billing Amount]) — Calculated a total of **$1.40bn** in gross billing.
- **ER Admission Rate:** Used DIVIDE and CALCULATE to determine that **32.93%** of all admissions are Emergency-based.
- **Average Length of Stay:** AVERAGE(Healthcare_dataset[Length of Stay]) — Identified a hospital-wide average stay of **15 days**.
- **Performance Tracking:** Implemented Top-N filtering and conditional formatting to rank the highest-grossing doctors and hospitals.

Business Impact & Stakeholder Value

The dashboard is divided into three functional layers to serve different hospital stakeholders:

1. **Executive Suite:** Provides a "Pulse Check" on Revenue ($1.40bn) and Admission Trends to manage high-level hospital growth.
2. **Clinical Heads:** Identifies "Abnormal" test result patterns across medical conditions (e.g., Cancer, Diabetes) to improve patient care quality.
3. **Operations Managers:** Enables deep-dives into Doctor and Hospital performance, identifying bottlenecks in 'Length of Stay' and 'Billing Efficiency'.